



Uganda Martyrs
University

UMU Attendance System

How it works — explained in everyday language

Uganda Martyrs University · Nkozi Campus

For Students, Lecturers, Faculty Admins, System Admins — and everyone else

What is this system?

It is a simple online register for classes. Instead of signing a paper sheet or calling out names, the lecturer shows a short code on screen, students type it into their phones within a few minutes, and the system records who was there.

- **No more paper registers** that get lost or filled in by friends.
- **Location check:** for physical classes, your phone must show you are on campus (or near the lecturer) to check in.
- **Automatic closing:** every class session closes itself when the class time ends — nobody has to remember to press "close".
- **Instant reports:** attendance per student, per course unit, per lecturer and per programme can be downloaded as PDF any time.
- **Email alerts:** students whose attendance falls below the rule are flagged so help can come early.

A class in 5 steps

1. Lecturer taps "**Open Session**" and picks the course unit and class length (e.g. 1 hour).
2. The app shows a **6-letter code** on the projector/screen.
3. Students open the app → the live class appears under "**Live Now**" → they type the code and tap **Check In**.
4. The code stops working after the chosen window (15–30 minutes); latecomers within the window still get in.
5. When class time ends, the session **closes automatically**. Anyone who never checked in is marked Absent. Lecturers can fix genuine mistakes.

Signing in — important to know

Email + password today is temporary. During building and testing we use plain email-and-password accounts so staff can be created quickly from CSV lists. This is only for the development phase.

In the final system, everybody signs in with their UMU Google account (Google sign-in). One button, no passwords to remember or steal. Works the same for students, lecturers and administrators, using your official UMU email address. First-time users just pick their faculty/programme once.

	Now (development)	Final system (production)
Students	Student email + password	Google sign-in (@stud.umu.ac.ug)
Staff	Staff email + password	Google sign-in (@umu.ac.ug)
Passwords stored?	Yes (temporary)	Not needed at all — Google handles it

The four roles at a glance

Role	What they do in one sentence
Student	Checks in with the code, sees own attendance and warnings.
Lecturer	Opens/closes classes, sees live names coming in, downloads reports.
Faculty Admin	Manages people and course units for one faculty; watches overall attendance.
System Admin	Runs the whole university setup: faculties, programmes, users, settings, audit log.

Questions students ask

Q: How do I mark myself present?

A: When your lecturer opens the class, it appears on your dashboard under "Live Now". Tap Check In, type the 6-letter code shown on the screen, and confirm. For physical classes keep location ON — the phone must say you are on campus.

Q: I forgot my phone / my battery died. What now?

A: Tell the lecturer. They can correct the record after class, and the change is logged with their name.

Q: The internet is slow — will I miss my mark?

A: The code stays valid for the whole window the lecturer chose (at least 15 minutes). As long as you check in before it expires you are counted present.

Q: Can a friend check in for me?

A: No. Each phone must also pass the campus/nearby-lecturer location check, and every correction a lecturer makes is recorded. Pretending to attend is an academic offence — not worth it.

Q: Why does my unit say "No sessions" instead of a percentage?

A: Because no class has been closed yet for that unit. Real percentages appear after the first session closes — we removed misleading numbers like "100%" when nothing has happened yet.

Q: Where do I see my history?

A: Dashboard shows this week and each unit's percentage; "View full attendance report" opens every past class, and Faculty Admins can print a PDF of it.

Questions lecturers ask

Q: How do I start a class?

A: Tap "Open Session", choose your course unit, set how long the class runs and how long the code stays valid, then open it. The code appears big on screen for the projector.

Q: What if I forget to close the session?

A: You cannot forget anymore. Every session now closes itself exactly when the class time you chose runs out, and absentees are recorded automatically.

Q: A student came late but within the code window — present or absent?

A: Present. The code window (15–30 min) is the grace period. You choose it when opening the class.

Q: Class ran longer than planned — can I give more time?

A: Yes, tap Extend on the live screen before the class ends; it adds time to both the code and the class clock. Near the end of class the system politely blocks extending and asks you to close instead.

Q: How do I prove attendance for a disputed case?

A: Open the session detail: every student, status and exact check-in time is listed, plus any edits made afterwards and who made them. You can download the report as PDF.

Q: Which reports can I download?

A: Per course unit (with charts), per student summary — as nicely branded UMU PDFs.

Questions Faculty Admins ask

Q: Who am I responsible for?

A: Only your own faculty. You see its students, lecturers, units, sessions and alerts. Other faculties are invisible to you; the System Admin sees everything.

Q: A course unit is missing — must I wait for the System Admin?

A: No. Use "Add a New Course Unit" on the Course Units page: type the name and code, and it lands straight in your faculty and appears in all dropdowns.

Q: What is the right order for importing CSV files?

A: Faculties → Programmes → Course Units → Curriculum → Staff → Students. The import screen now shows this order with green ticks and warns you when a file depends on codes you have not imported yet, and checks your file's columns before uploading.

Q: How do I spot struggling students early?

A: Your dashboard groups students into Warning and Critical based on attendance rules, with a per-student list you can work through.

Q: Can I fix a wrong attendance mark?

A: Lecturers handle corrections for their sessions (each edit is audit-logged). You have full visibility of the records in your faculty.

Q: Can I share a unit taught in two faculties?

A: Units can be shared with additional faculties; the System Admin sets that sharing, you see it reflected in your lists.

Questions System Admins ask

Q: What can I do that others cannot?

A: Create/edit faculties and programmes, import staff and students in bulk, set the global academic year and semester, manage every user account, adjust global settings, read the full audit log, and trigger system updates.

Q: How do accounts get created without Google sign-in yet?

A: CSV imports create accounts with a default password; users are forced to change it at first login. Names and real identities arrive from Google on first sign-in once Google Auth is switched on for everyone.

Q: Is the data safe?

A: Access is limited strictly by role, sensitive actions are written to an audit log, traffic is encrypted with HTTPS, and the database is backed up regularly. With Google sign-in, no passwords are stored here at all.

Q: Where does the system run?

A: One Ubuntu server running everything as Docker containers: the web app, MySQL database, Redis (speed & protection against flooding), and Nginx (front door with HTTPS). Updates are one command away.

Q: What if two copies of the app run at once?

A: Safe by design: the auto-close scheduler elects a leader through Redis, and closing a session twice is impossible thanks to guarded database updates.

Q: How does the campus location rule work?

A: Physical check-ins compare the phone's GPS position against the campus boundary and, where configured, the lecturer's position within a radius. Online sessions skip location entirely.

Questions technical people ask

Q: What is it built with?

A: Web client: React + TypeScript (works offline-ish, installable as an app). Server: Node.js/Express with Prisma. Database: MySQL 8. Extras: Redis for caching/rate-limiting/leader election, Puppeteer for PDF generation, Nginx reverse proxy, all deployed with Docker Compose.

Q: How does authentication work end-to-end?

A: Development phase: email + hashed password (seeded default forced to rotate). Production plan: Google OAuth for ALL users — the backend verifies the Google identity and maps it to the user record; no local passwords remain.

Q: How does auto-close actually work?

A: A scheduler ticks every minute, finds open sessions whose openedAt + chosen duration has passed, atomically closes them (status-guarded update), writes absent records for non-check-ins, and logs an audit event. Leader election prevents duplicate work across replicas.

Q: How are reports produced?

A: Each report endpoint assembles header + tables, wraps them in a branded HTML template and prints it to A4 PDF via headless Chrome (Puppeteer). Four generators exist: lecturer, programme, course-unit and student.

Q: What protects against brute-force or floods?

A: Rate limiting backed by Redis on sensitive endpoints, unique 6-character session codes from a safe alphabet with short TTL, and role-scoped authorization on every route.

Q: How is it updated?

A: git push → pull on the server → rebuild containers → prisma migrate deploy. A built-in "Update System" button runs the same pipeline and streams the log back to the admin.

One-page summary

Feature	In plain words
Check-in code	Six letters on the screen; type them in the app before the window closes.
Campus rule	Physical classes require your phone to place you on campus / near the lecturer.
Auto-close	Sessions always end by themselves after the chosen class length — no option to leave one open forever.
This week card	Shows Monday–Sunday of the current week, refreshed live.
Honest stats	Until a first class closes, units show "No sessions" instead of a fake percentage.
Easy setup	Guided import order, column pre-checks, and admins can add missing units by hand.
Curriculum tables	One editable table per programme: add/remove units per year and semester inline.
PDF reports	Branded downloadable reports for lecturers, programmes, units and students.
Login	Email+password exists ONLY during development; production uses UMU Google sign-in for everyone.
Fairness	Every change to a record is logged with who made it and when.

Bottom line: honest attendance, less paperwork, early warning for struggling students — and a login that will feel as familiar as checking your university email.